

Course Project Overview

In this course project, you will carry out a significant design project using methods and techniques you will learn in the HCI & Computer Graphics course. Through a series of activities, marked off by Milestones, you will carry your project through from an initial concept to a finished prototype.

Your project will be in response to a *design challenge*, **a description of a problem that your design will solve**. You have been provided two design challenges from which you can choose. These challenges are fairly broad, allowing you latitude in choosing your specific problem, intended user population, target platform(s), etc. You can also propose your own design challenge, some guidance is provided on how and why to choose this route in the “Create Your Own” Challenge below.

Regardless of the challenge you choose, you will follow the same process, produce the same deliverables, and meet the same criteria. In either case you will be **designing a website or mobile app** (while it’s possible to design something that can work as both, to keep the scope manageable it’s best to focus on one platform for this project). We will refer to the website or app that you are designing as your “product” or your “system” in subsequent descriptions. Since you won’t be (almost) actually programming your system, you don’t need to worry about technical details such as which particular platforms, programming languages, or technologies will be used to produce your design. You can assume capabilities that are more or less standard in contemporary web and mobile apps, such as dynamic data-driven interfaces, some degree of personalization, and multimedia capabilities.

So, let’s dive in! Check out the design challenges below, and start thinking about what you’d like to do for your project!

Design Challenge 1: Productivity and Organization

Modern life is busy. Many people are stressed out trying to balance work, school, family, and a social life. We’re all trying to improve ourselves at the same time. Get exercise, eat healthy, learn new skills, contribute to the community... It’s overwhelming! Technology often makes things worse (in terms of being overwhelmed), but in this challenge you will design a tool that *uses technology to help people stay organized and/or productive* in the ways that matter to them. Choose a *user population* and an *activity or family of activities* where you think technology could help, then work through the design process to understand user needs, generate design concepts, prototype and test, and come out with a design for a product that could help *those users* more effectively *manage or perform those activities*.

There are many, many directions you could take with this. Here are a few examples:

- Tracking progress and providing motivation towards personal health or fitness goals (e.g., exercise or nutrition)
- Managing tasks and to-dos, whether for the individual or for groups working towards a common goal
- Coordinating activities within a family/household: meal planning, kids activities,

household chores, family togetherness

- Providing support for collaboration and group activities: group scheduling for work, social, or other activities

And many more... There are many, many possibilities for where you could take this.

Some tips to get started:

- Think about people close to you: friends, family, co-workers, people in your community. Where could you design something to solve a real problem and provide real benefit?
- Think about real people: you will conduct a study to learn more about user needs, so you don't need to know *everything* about your users at the outset (in fact you can't!) but without somewhere to start, you won't know who to ask or what to ask them! Products designed "for everyone" are usually not good for anyone. Be specific about who your users are and what they need to do. The more specific you can be, the better and more unique your design is likely to be in the end.
- Look around on the Web, app stores, etc. to see what's already out there. You don't have to convince yourself (or us) that there is *nothing* out there that would address your users' need, but you should be at least somewhat convinced that you can come up with something that could offer some advantages in at least some situations.

Design Challenge 2: Community Technology

Humans are social animals. We gather together into families, communities, and groups with shared interests. Many think that technology drives us apart, we have less need to interact with people in our physical environment, we are more satisfied by digital interactions than interpersonal interactions, we have more opportunities to connect with faraway strangers than we do to connect with our neighbors. In this challenge you will design a tool that *uses technology to help strengthen communities* by allowing members of those communities to connect, share, and/or work together. Choose a *community* and an area where you think technology could help address a *community need*, then work through the design process to understand user needs, generate design concepts, prototype and test, and come out with a design for a product that could help *your selected community* (or type of community) address the *particular needs you have identified*.

There are many, many directions you could take with this. Here are a few examples:

- Tools for sharing information and resources within a community, for example neighborhood watch programs, or borrowing/lending networks
- Something that would help your local government work more smoothly for particular members of the populace or for particular communities, for example, better information sharing between citizens and government, tools for reporting problems and tracking progress on attempts to fix them.
- Community organizing, tools for helping members of a community to gather and communicate support for issues that are important to them as a way to influence decision - making
- Building connections, tools that would help people get to know their neighbors better, and differently than they might through other means.

Some tips to get started:

- Think about communities you are a part of: your neighborhood, your city or town, your school, your workplace, your mosque/temple/church, any groups or

organizations you belong to.

- In some cases, you may be able to talk to people who are in charge of the communities of interest, city council members, group leaders, work managers, school administrators, and find out about needs or ideas that they already have in mind. Forming partnerships like this could be very helpful when it comes time to recruit participants for interviews and user testing.
- Look around on the Web, app stores, etc. to see what's already out there. You don't have to convince yourself (or us) that there is *nothing* out there that would address your users' needs, but you should be at least somewhat convinced that you can come up with something that could offer some advantages in at least some situations.
- Think about real people: you will conduct a study to learn more about user needs, so you don't need to know *everything* about your users at the outset (in fact you can't!) but without somewhere to start, you won't know who to ask or what to ask them! Products designed "for everyone" are usually not good for anyone. Be specific about who your users are and what they need to do. The more specific you can be, the better and more unique your design is likely to be in the end.

Design Challenge 3: Create your own!

If you already have an idea for an app/site/service that you've been dying to work on, this could be your chance. You'll still have to follow the process and milestones laid out for the course, but you can work on a problem and project of your own choosing. To take this route you'll have to clearly define your target population, the need(s) to be addressed, the way your system will address those needs, and why you think your system will be better than what your users are doing now. A good idea to consider is whether you might be able to find a "client" for your project, i.e., an organization that would like to have a website/app designed, especially if they don't have the resources or skills to be able to do it themselves. Nonprofits, small companies, social groups, etc. could be great candidates, and you would benefit from having help defining a compelling problem, identifying and recruiting users, and getting feedback along the way!

Some tips to get started:

- Think about your target users. Do you know anybody personally who would be in that category? If not, it might be an indication that this project will not be a good fit for the course project. You will need to make contact with potential users several times throughout the project, so being able to access a good number of them easily will be an important key to success.
- Be as specific as possible about the user needs you will seek to address. Having a concrete, coherent focus is essential for keeping a project on track and well-scoped.
- Look around on the Web, app stores, etc. to see what's already out there. You don't have to convince yourself (or us) that there is *nothing* out there that would address your users' need, but you should be at least somewhat convinced that you can come up with something that could offer some advantages in at least some situations.

Milestone 0 - Submission

Defining Your Project

Starting from one of the challenges described here, you will have to do some work (mainly thinking, reflection, and some lightweight research) to more clearly nail down a specific user population and a problem that you want to solve for that population. This is one of the key activities in Milestone 1, and a bit more detail is provided there about what to do and how to develop your *project brief*, which is short document that describes the project you plan to do. As you work through different ideas, feel free to reach out to your friends or the instructor(s) for feedback, inspiration, and questions if you feel you need clarification.

To get you started, you must work in your teams to brainstorm, ideate and come up with at least 3 design challenges. Submit this along with your proposed team at:

<https://forms.gle/aojtEVA7yip9jwGD7>

Defining Your Team

You will be doing this project in a group (team of 3 - 4). As the first step, submit the project registration form given here at this link: <https://forms.gle/aojtEVA7yip9jwGD7>

Note: The design challenge you select for your project must be unique, especially in terms of what you have already worked on. You are not allowed to select your course project from a previous course you have taken in the degree as your design challenge. If you happen to work on something similar, then make sure to clearly justify the specific 'need' that you will cater for which remains unaddressed in your previous undertaking of the project. You must discuss this with the instructor and seek approval.

Submission Deadline:

Please refer to google classroom.